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Cite as: Appl. Phys. Lett. **115**, 221103 (2019); <https://doi.org/10.1063/1.5124735>

Submitted: 18 August 2019 . Accepted: 07 November 2019 . Published Online: 26 November 2019

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ABSTRACT

All-solution-processed quantum dot (QD) upconversion devices were realized by constructing a colloidal PbS QD near-infrared (NIR $\sim 1 \mu\text{m}$)-sensitive layer on the cathode side of a red CdSe quantum dot light-emitting diode (QLED). Different from conventional upconversion devices, we utilized the photogenerated electrons rather than the holes created in the NIR-sensitive layer to inject into the QLED for visible light emission. Consequently, we obtained high-performance devices with a maximum photon-to-photon conversion efficiency of 3.35% and a luminance on-off ratio of approximately 8×10^3 . The results indicate that the device structure could be used to realize low-cost and portable high-performance QD-based upconversion devices via solution-processing.

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Near-infrared (NIR)-to-visible upconversion devices have received considerable attention recently due to their potential applications in the NIR imaging field, including medical imaging, production monitoring, night vision, and defense security.^{1–4} As the name implies, NIR-to-visible upconversion devices provide an effective approach to convert incident NIR light to output visible light. In the past few decades, there have been increased research efforts to develop various materials and device architectures for NIR-to-visible upconversion devices. Typically, an NIR-to-visible upconversion device is created by integrating a photodetector (PD) and a light-emitting diode (LED). In these devices, the LED turns on and emits visible light originating from the NIR light-induced photocurrent by the PD, which offers a low-cost and pixel-free infrared imaging solution.^{2,5} For NIR-to-visible upconversion devices, various infrared absorbing and luminescent materials, including inorganic,⁶ organic,⁷ organic/inorganic hybrid,² PbSe (PbS) quantum dots (QDs),^{8–11} inorganic perovskite, and organic/inorganic hybrid perovskite, have all been used as NIR absorbers and visible light-emitting layers.^{12,13} Despite fast development and

a noticeable improvement in upconversion performance for NIR-to-visible upconversion devices,^{14,15} they still suffer from lattice mismatch,^{2,16} short-wavelength detection spectra,¹⁷ low color saturation, and incompatible fabrication processes.^{2,13,18}

In recent years, great effort has been undertaken to address the aforementioned issues. In 2011,⁸ PbSe QDs used as NIR-sensitive layers were the first reported NIR-to-visible upconversion devices with QD materials. Facile solution-process methods for PbSe QD films can facilitate the achievement of cost reduction for upconversion device fabrication. Particularly, since the optical absorption of PbSe (PbS) QDs can be tuned from 0.4 to 2.0 μm , these hybrid upconversion devices show a broad detection spectrum (from the visible region to 1.5 μm).^{8,9} However, the 6 V turn-on voltage (V_{on}) was quite high, and the photon-to-photon conversion efficiency of devices is still less than expected. With the development of perovskite optoelectronic materials and devices, perovskite/organic hybrid upconversion devices have also been investigated to improve device performance. Yu and co-workers first reported an NIR-to-visible upconversion device using

a hybrid perovskite as an NIR absorber integrated with phosphorescent organic light-emitting diodes (OLEDs); the device showed a rather low V_{on} of 1.9 V and an NIR-to-visible upconversion efficiency of 3%.¹² Nevertheless, the spectral detection range of these upconversion devices is 800–900 nm, which is much shorter and narrower than that of upconversion devices with PbSe or PbS QDs as the NIR-sensitive layer.

Recently, an NIR-to-visible upconversion device based on an organic bulk heterojunction NIR charge generation layer (CGL) and a CsPbBr₃ perovskite LED unit has been reported.¹³ Although using a solution-processable CsPbBr₃ perovskite as a light-emitting layer achieved a saturated 520-nm green light electroluminescence (EL) emission, the blend of diketopyrrolopyrrole-dithienylthieno[3,2-b]thiophene (DPP-DTT) polymer with a nonfullerene acceptor CO₈DFIC employed as a photosensitive layer suppressed the NIR light absorption, leading to photoresponsive reduction in the long-wavelength range. Furthermore, a 100-nm-thick TPBi (2,2',2''-(1,3,5-Benzinetriyl)-tris(1-phenyl-1-H-benzimidazole)) electron-transporting layer (ETL) in the perovskite LED part of the upconversion device was deposited by thermal evaporation in a vacuum chamber. Thus, this did not fulfill the criteria of an all-solution-processable device due to the increased complexity of the fabrication process.

In the meantime, several recent research groups have reported nanoparticles and solid-state thin films that directly upconvert IR without requiring an applied bias voltage,^{19–21} but the optical upconversion is a process that converts two or more low-energy photons into a single high-energy photon that is usually associated with low efficiency.¹⁹ However, electrically driven upconversion devices enjoy much higher photon-to-photon upconversion efficiency up to 100%.¹⁴ The efficiency can be even much higher than 100% by incorporating a phototransistor or photomultiplying organic NIR photodetector with the light-emitting device.^{11,14} Associated with the efficiency, electrically driven upconversion devices can operate at lower NIR illumination compared to optical ones. Nevertheless, the fabrication processes and properties of NIR-to-visible upconversion devices are far from satisfactory.

PbS QDs are a kind of ambipolar semiconductor material that exhibits significant ambipolar transport properties for both holes and electrons. According to previous studies,^{22,23} balanced carrier mobility is difficult to achieve in PbS QDs. Shulga *et al.* reported double-gate transistors based on PbS QDs as the active channel layer,²⁴ the electron mobility value of the PbS QDs was much higher than the hole mobility (by more than two orders of magnitude) when using either SiO₂ or a polymer as the gate dielectric material. Recently, other studies on PbS QDs have demonstrated similar trends, which implies that electron transport is more efficient than hole transport in optoelectronic devices with PbS QDs.^{25,26} Nevertheless, for NIR-to-visible upconversion devices with a PbS QD CGL, only photogenerated hole-based upconversion devices have been reported until now. On the other hand, despite the unique characteristics that tunable narrow linewidth emission,²⁷ high color saturation,²⁸ high EL efficiency, and solution-process fabrication have been achieved in quantum dot light-emitting diodes (QLEDs),^{29,30} NIR-to-visible upconversion devices based on QD light-emitting layers have not yet been explored.

In this work, we realized high-performance solution-processed QD NIR-to-visible upconversion devices by integrating a CdSe/ZnS QLED and utilizing photogenerated electrons from the PbS QD NIR absorber layer. All of the components in the devices were fabricated

via solution-processing except for the anode and cathode. They showed a luminance on-off ratio as high as 8×10^3 when under incident NIR light illumination and upconversion efficiency up to 3.35% at 970 nm. This indicates that the high performance is mainly derived from the highly efficient transportation of photogenerated electrons and their strong potential to achieve a balance between the holes and electrons in the device. Additionally, the NIR-to-visible upconversion devices emitted well saturated red light at 628 nm. The results indicate that the QD-based upconversion device not only can be applied to investigate the transport mechanism of the photogenerated electrons originated from the PbS QDs in the NIR-to-visible upconversion devices but also offers a prospective opportunity to simplify the fabrication process, lower the research cost, and develop high-performance NIR-to-visible upconversion devices.

As shown in Fig. 1(a), an upconversion device with the structure ITO/PEDOT:PSS/TFB/CdSe/ZnS QDs/PbS QDs/Al was fabricated by solution-processing. The CdSe/ZnS QDs and PbS QDs were used as the emission layer (EML) and CGL, respectively. The other layers are typical constituents of a QLED, among which ITO works as the anode and PEDOT:PSS [poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate)], TFB [poly((9,9-dioctylfluorenyl-2,7-diyl)-co-(4,4'-(N-(4-s-butylphenyl)diphenyl-amine))], and ZnO act as the hole injection layer (HIL), hole transport layer (HTL), and ETL, respectively. The detailed experimental procedures for the PbS QD synthesis and device fabrication are described in the [supplementary material](#).

Figure 1(b) shows the detailed band structure of the designed device, while Figs. 1(c) and 1(d) depict the operational mechanism of NIR-to-visible upconversion devices. In a dark environment when there is no NIR illumination, electrons injected into the Al electrodes are blocked due to the electron barrier between the Al cathode and the PbS QD CGL [Fig. 1(c)]. On the other hand, holes are injected into the ITO anode and accumulate at the EML/ETL interface between the CdSe QD and ZnO nanocrystal layers because of a high hole barrier.

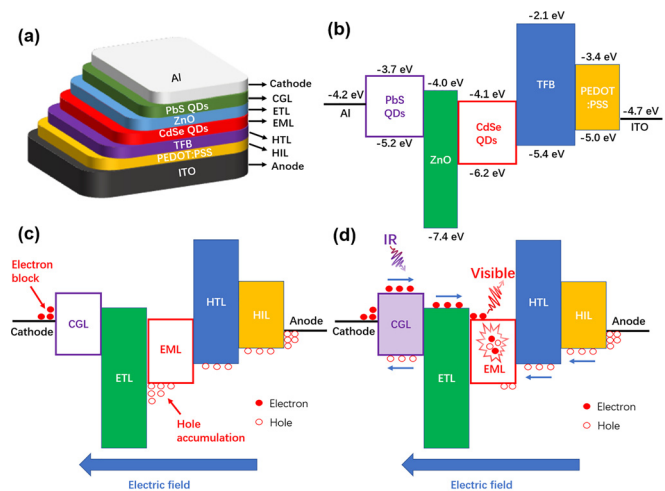


Figure 1

FIG. 1. Schematic representation of (a) the layers in the NIR-to-visible upconversion device structure, (b) an energy-band diagram of the materials and the device configuration, and the corresponding working mechanisms (c) without and (d) with NIR illumination.

Hence, there is no light emission in the device. When NIR light impinges on the device, it is absorbed in PbS QDs creating photogenerated excitons, as shown in Fig. 1(d). The photogenerated excitons are dissociated into electrons and holes by the external electric field. The photogenerated electrons are injected into the EML through the ETL and recombine with the holes originating from the ITO anode, resulting in visible light in the QLED unit. In this case, it is worth mentioning that the PbS QDs also function by blocking the electrons from the cathode. Thereby, the incident NIR light can be converted to output visible light in the NIR-to-visible upconversion devices.

Figure 2(a) presents the current and luminance characteristics of the upconversion devices with and without NIR illumination. The dark current density of the upconversion devices without NIR illumination was reduced by ~ 3 times than with it, suggesting that NIR illumination effectively increased the current density of the device from the photogenerated charge in the CGL. Nevertheless, the dark current still flows through the upconversion device under a higher applied voltage because this drives the electrons over the barrier, which generates the dark current. In the absence of NIR power, the low but slightly increased dark current mainly originates from the large electron barrier between the PbS QD layer and the Al cathode, thereby hindering the electron injection (similar results can be found in the literature).¹⁸

In addition, the 150-nm-thick PbS QD film also increases the series resistance of the upconversion device. Thus, V_{on} was high up to 5.7 V under no NIR illumination (the turn-on luminance was 10^{-2} cd/m² for this upconversion device). On the other hand, under illumination (4.73 mW/cm²), V_{on} of the upconversion device was as low as 2.5 V and the luminance increased rapidly and reached 330 cd/m² at 10 V. By comparing the data with and without NIR excitation, the sharp luminance demonstrates that photogenerated electrons in the PbS QD layer can efficiently transfer into the EML through the ETL because there is no electron barrier between the CGL and the ETL. Above 6 V, the luminance of the upconversion device became nearly saturated, which implies that luminance is restricted by the transport supply of the photogenerated charge in the PbS QD layer.¹²

The current density-voltage and luminance-NIR intensity characteristics of the upconversion devices at different NIR power intensities and applied bias voltages are shown in Figs. 2(b) and 2(c). In contrast, the current density and luminance increase gradually as the NIR power density enhances. Similar results have been reported in the all-organic upconversion devices.¹⁸ Increasing the illumination power density can increase the number of photogenerated charges in the PbS QDs, thus increasing the current density of the upconversion devices. We should note that the current density and luminance of these QD upconversion devices based on the PbS QD NIR-sensitive layer are observably higher than those of organic upconversion devices based on the PbS (PbSe) QD absorber. In previous studies on PbS (PbSe) QDs, only photogenerated holes were utilized to inject into the light EML in the hybrid organic-QD upconversion devices.^{8,9} However, the mobility of the electrons is obviously superior to that of the holes in the PbS QDs.^{24–26} In our case, the photogenerated electrons that emerged in the PbS QDs under NIR illumination were efficiently injected into the QLED units of the QD upconversion devices under the external electric field. Therefore, high brightness and current density in the QD upconversion devices are primarily due to efficient transport of photogenerated electrons.

To illustrate the switching property of the QD upconversion devices, the luminance on-off ratios of the devices at different NIR

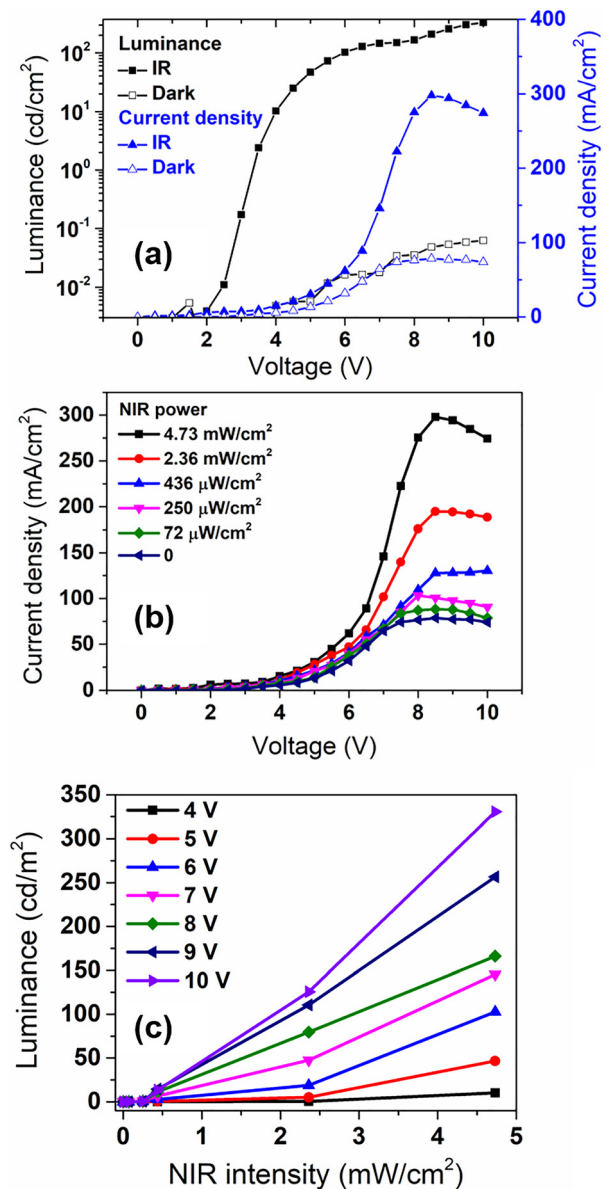


FIG. 2. The current density and light emission characteristics of the NIR-to-visible upconversion devices: (a) luminance-current density-voltage with and without incident NIR light, (b) current density as a function of voltage at different NIR intensities, and (c) luminance as a function of NIR intensity for different voltages.

intensities are shown in Fig. 3(a). With increasing applied NIR power density, the luminance on-off ratio of the devices increased by two orders of magnitude. In particular, for a 7 V bias, the luminance on-off ratio reached 8×10^3 , which is distinguishable by the naked eye. When under NIR illumination, the CGL layer in the upconversion devices supplies an additional voltage to the QLED unit serving as a photovoltaic cell. To better elucidate the photovoltaic effect in the upconversion devices, the turn-on voltages as a function of NIR intensity are shown in Fig. 3(b). When the NIR intensity increased, the

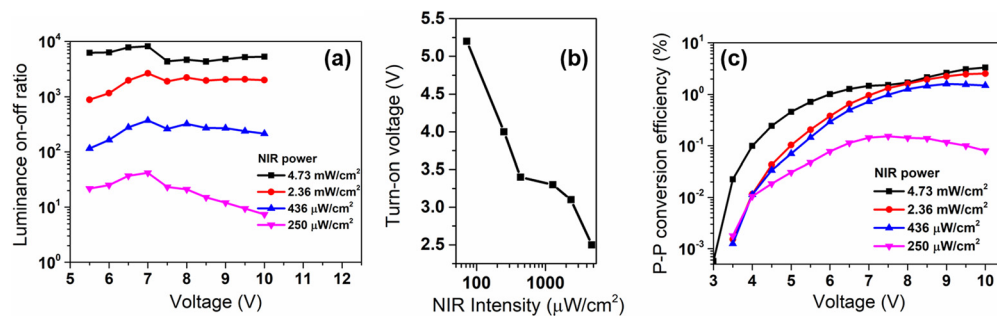


FIG. 3. (a) On-off ratio of luminance at different NIR power densities, (b) turn-on voltage as a function of NIR intensity, and (c) P-P conversion efficiency at different NIR power densities of the upconversion devices.

turn-on voltage of the devices gradually decreased from 5.2 V to 2.5 V. Yu *et al.* used the hybrid perovskite as an NIR absorber in NIR-to-visible upconversion devices;¹² their research showed that the FAPbI₃ layer functions as a photovoltaic cell that reduced the turn-on voltage. These results demonstrate that the photovoltaic effect reduces the turn-on voltage of the LED unit at high NIR power density. Based on the calculation,¹⁴ the upconversion efficiency reached over 3% [Fig. 3(c)], which is higher than that of PbS (PbSe) QD-based upconversion devices harvesting photogenerated holes.^{8–10} This is due to the fact that the harvesting photogenerated electrons are efficiently injected into the EML.

To investigate the electroluminescence (EL) spectrum distribution of the upconversion devices, we measured their performances for various driving voltages at different NIR irradiation intensities, as shown in Figs. S1(a)–S1(c). In addition, we compared the CIE coordinates of

the QD-based upconversion devices in this work with those of the previously reported ones (Fig. 4). The red EL emission with a narrow full-width at half-maximum (FWHM) of 20–24 nm corresponds to CIE 1931 color coordinates of (0.68, 0.31), which are very close to the CIE coordinates of a high-performance red light QLED.³¹ The color saturation property makes this red upconversion device a promising choice for applications in the NIR imaging field.

In conclusion, we fabricated a class of solution-processed QD NIR-to-visible upconversion devices based on a CdSe/ZnS QD EML and a PbS QD NIR-sensitive CGL. Our results indicate that these layers are competent for the upconversion process due to their unique virtues of narrow linewidth emission, high color saturation, and broad sensitivity spectrum. The maximum photon-to-photon conversion efficiency at a peak wavelength of 970 nm reached over 3%. The luminance on-off ratio of the optimized devices reached 8×10^3 , thereby demonstrating excellent switching characteristics. The results of this study reveal that both QD materials are excellent choices for the fabricated all-solution-processed upconversion devices, which could lead to the development of high-performance infrared imaging devices with a long lifetime via simple fabrication at low-cost.

See the [supplementary material](#) for the experimental section, details of EL spectra distributions (Fig. S1), absorption and photoluminescence (PL) spectra of PbS QDs and CdSe/ZnS QDs in solution, a TEM image of PbS QDs (Fig. S2), a photo of the NIR-to-visible upconversion device (Fig. S3), and the on-off behavior of the upconversion device without and with NIR in the video.

This work was supported by the National Key Research and Development Program of China administrated by the Ministry of Science and Technology of China (Nos. 2016YFB0401702 and 2017YFE0120400), National Natural Science Foundation of China (Nos. 61674074, 61405089, 61704170, and 61875082), Development and Reform Commission of Shenzhen Project (No. [2017]1395), Shenzhen Peacock Team Project (No. KQTD2016030111203005), Shenzhen Key Laboratory for Advanced quantum dot Displays and Lighting (No. ZDSYS201707281632549), Guangdong University Key Laboratory for Advanced Quantum Dot Displays and Lighting (No. 2017KSYS007), Distinguished Young Scholar of National Natural Science Foundation of Guangdong (No. 2017B030306010), and Shenzhen Innovation Project (No. JSGG20170823160757004) and also thanks the start-up fund from Southern University of Science and Technology.

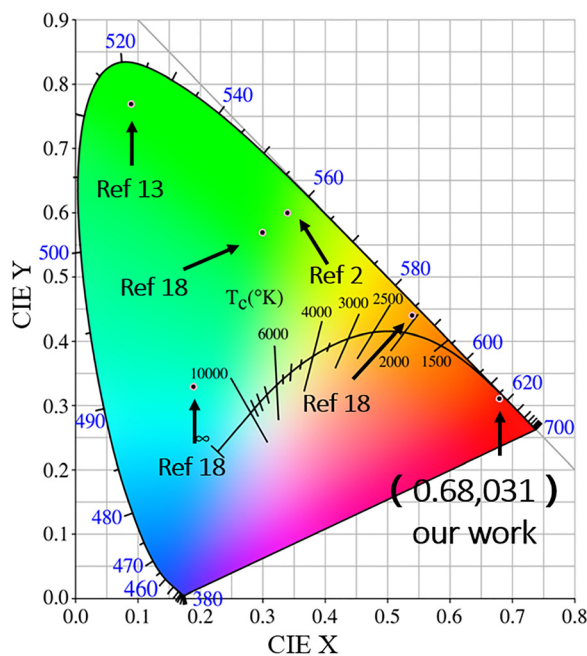


FIG. 4. CIE coordinates of the EL spectra for the upconversion device in this work and others reported in the literature.

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